

第一章 分散系

1.1 (1) 溶液的凝固点下降

(2) 土壤溶液浓度过大, 渗透压大于植物根细胞液的渗透压

(3) 溶液的凝固点下降

1.2 沸点不断上升, 至溶液达到饱和后, 沸点恒定;

蒸气凝结温度恒定, 等于溶剂的沸点。

1.3

$$c(\text{H}_2\text{O}_2) = \frac{w(\text{H}_2\text{O}_2)\rho}{M(\text{H}_2\text{O}_2)} = \frac{3.0\% \times 1000 \text{ g} \cdot \text{L}^{-1}}{34 \text{ g} \cdot \text{mol}^{-1}} = 0.88 \text{ mol} \cdot \text{L}^{-1}$$

$$b(\text{H}_2\text{O}_2) = \frac{w(\text{H}_2\text{O}_2)/M(\text{H}_2\text{O}_2)}{1 - w(\text{H}_2\text{O}_2)} = \frac{3.0\% / 0.034 \text{ kg} \cdot \text{mol}^{-1}}{97.0\%} = 0.91 \text{ mol} \cdot \text{kg}^{-1}$$

$$x(\text{H}_2\text{O}_2) = \frac{n(\text{H}_2\text{O}_2)}{n(\text{H}_2\text{O}_2) + n(\text{H}_2\text{O})} = \frac{b(\text{H}_2\text{O}_2)}{b(\text{H}_2\text{O}_2) + 1/M(\text{H}_2\text{O})} = 1.6\%$$

1.4 凝固点自高至低顺序: 葡萄糖溶液、醋酸溶液、氯化钾溶液

1.5

$$b = 1.17 \text{ mol} \cdot \text{kg}^{-1}$$

$$\Delta T_b = K_b b = 0.52 \text{ K} \cdot \text{kg} \cdot \text{mol}^{-1} \times 1.17 \text{ mol} \cdot \text{kg}^{-1} = 0.61 \text{ K}$$

$$T_b = 373.76 \text{ K} = 100.61^\circ \text{C}$$

$$\Delta T_f = K_f b = 1.86 \text{ K} \cdot \text{kg} \cdot \text{mol}^{-1} \times 1.17 \text{ mol} \cdot \text{kg}^{-1} = 2.18 \text{ K}$$

$$T_f = 270.87 \text{ K} = -2.18^\circ \text{C}$$

1.6 $\pi = cRT = \frac{m/M}{V} RT$

$$M = \frac{mRT}{V\pi} = \frac{0.40 \text{ g} \times 8.31 \text{ kPa} \cdot \text{L} \cdot \text{mol}^{-1} \cdot \text{K}^{-1} \times 300 \text{ K}}{0.10 \text{ L} \times 0.499 \text{ kPa}} = 2.0 \times 10^4 \text{ g} \cdot \text{mol}^{-1}$$

1.7

$$\frac{9.5}{1} : \frac{10.2}{16} : \frac{80.3}{12} = 30 : 2 : 21$$

化合物中 C、H、O 原子数比为 21: 30: 2

$$M = \frac{K_f m_B}{\Delta T_b m_A} = \frac{5.12 \text{ K} \cdot \text{kg} \cdot \text{mol}^{-1} \times 0.100 \text{ g}}{0.33 \text{ K} \times 5.00 \text{ g}} = 310 \text{ g} \cdot \text{mol}^{-1}$$

故该化合物的化学式为 $\text{C}_{21}\text{H}_{30}\text{O}_2$

1.8

$$\frac{m\{(\text{NH}_2)_2\text{CO}\} / M\{(\text{NH}_2)_2\text{CO}\}}{m(\text{H}_2\text{O})} = \frac{m_B / M_B}{m(\text{H}_2\text{O})}$$

$$\therefore M_B = 342.4 \text{ g} \cdot \text{mol}^{-1}$$

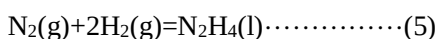
第二章 化学热力学基础

2.1 (1)错误; (2)正确; (3) 错误; (4) 错误; (5)正确; (6)正确; (7) 错误; (8) 错误

2.2 (1/4) [反应式(1) - 反应式(2)]得: $(1/2)\text{N}_2(\text{g}) + (1/2)\text{O}_2(\text{g}) = \text{NO}(\text{g})$

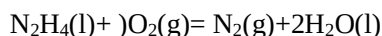
$$\begin{aligned}\therefore \Delta_f H_m^\theta(\text{NO}, \text{g}) &= (1/4) \{ \Delta_r H_m^\theta(1) - \Delta_r H_m^\theta(2) \} \\ &= (1/4) [-1107 \text{ kJ} \cdot \text{mol}^{-1} - (-1150 \text{ kJ} \cdot \text{mol}^{-1})] \\ &= 90 \text{ kJ} \cdot \text{mol}^{-1}\end{aligned}$$

2.3 (1/4) [反应式(3) - 反应式(4) + 3 × 反应式(2) - 反应式(1)]得:



$$\begin{aligned}\therefore \Delta_f H_m^\theta(\text{N}_2\text{H}_4, \text{g}) &= (1/4) \{ \Delta_r H_m^\theta(3) - \Delta_r H_m^\theta(4) + 3 \times \Delta_r H_m^\theta(2) - \Delta_r H_m^\theta(1) \} = \\ &= (1/4) \{ -143 \text{ kJ} \cdot \text{mol}^{-1} - (-286 \text{ kJ} \cdot \text{mol}^{-1}) + 3 \times (-317 \text{ kJ} \cdot \text{mol}^{-1}) - (-1010 \text{ kJ} \cdot \text{mol}^{-1}) \} = \\ &= 50.5 \text{ kJ} \cdot \text{mol}^{-1}\end{aligned}$$

2 × 反应式(4) - 反应式(5) 得:

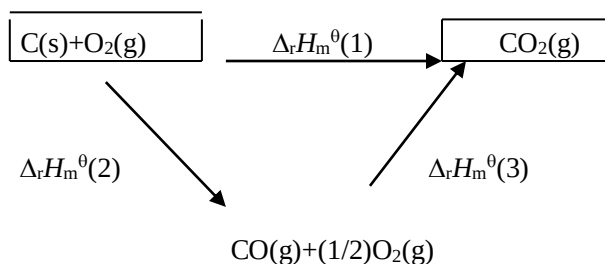


$$\Delta_r H_m^\theta = 2 \times \Delta_r H_m^\theta(4) - \Delta_r H_m^\theta(5) = 2 \times (-286 \text{ kJ} \cdot \text{mol}^{-1}) - 50.5 \text{ kJ} \cdot \text{mol}^{-1} = -622.5 \text{ kJ} \cdot \text{mol}^{-1}$$

2.4 $\Delta_r H_m^\theta = 2\Delta_f H_m^\theta(\text{CO}_2, \text{g}) + 3\Delta_f H_m^\theta(\text{H}_2\text{O}, \text{l}) + (-1) \times \Delta_f H_m^\theta(\text{CH}_3\text{OCH}_3, \text{l}) + (-3)\Delta_f H_m^\theta(\text{O}_2, \text{g})$

$$\therefore \Delta_f H_m^\theta(\text{CH}_3\text{OCH}_3, \text{l}) = 2\Delta_f H_m^\theta(\text{CO}_2, \text{g}) + 3\Delta_f H_m^\theta(\text{H}_2\text{O}, \text{l}) - \Delta_r H_m^\theta = -183 \text{ kJ} \cdot \text{mol}^{-1}$$

2.5



由题意知, $\Delta_r H_m^\theta(1) < 0$, $\Delta_r H_m^\theta(2) < 0$, $\Delta_r H_m^\theta(3) < 0$

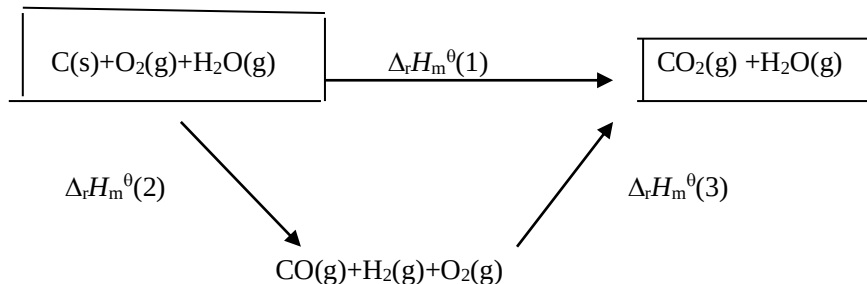
$$\therefore \Delta_r H_m^\theta(1) = \Delta_r H_m^\theta(2) + \Delta_r H_m^\theta(3)$$

$$\Delta_r H_m^\theta(1) - \Delta_r H_m^\theta(3) = \Delta_r H_m^\theta(2) < 0$$

即: 以碳直接作燃料时放热较多

2.6 $\text{C}(\text{s}) + \text{H}_2\text{O}(\text{g}) = \text{CO}(\text{g}) + \text{H}_2(\text{g})$

$$\begin{aligned}\Delta_r H_m^\theta &= \Delta_f H_m^\theta(\text{CO}, \text{g}) + (-1)\Delta_f H_m^\theta(\text{H}_2\text{O}, \text{g}) = \\ &= -110.5 \text{ kJ} \cdot \text{mol}^{-1} - (-241.8 \text{ kJ} \cdot \text{mol}^{-1}) = 131.3 \text{ kJ} \cdot \text{mol}^{-1}\end{aligned}$$



$$\therefore \Delta_r H_m^\theta(1) = \Delta_r H_m^\theta(2) + \Delta_r H_m^\theta(3) \quad \therefore \Delta_r H_m^\theta(1) - \Delta_r H_m^\theta(3) = \Delta_r H_m^\theta(2) > 0$$

由题意知, $\Delta_r H_m^\theta(1) < 0$, $\Delta_r H_m^\theta(3) < 0$ 故: 以水煤气作燃料时放热较多

2.7 均为熵增过程。

2.8 标准状态下:

$\Delta_r H_m^\theta$	$\Delta_r S_m^\theta$	
<0	<0	低温自发
<0	>0	任意温度均自发
>0	>0	高温自发
>0	<0	任意温度均不可能自发

2.9 (2) (反应为气体物质的量增加的过程, $\Delta_r S_m^\theta > 0$)

2.10 不可以。因为物质的 S_m^θ , 不是指由参考状态的元素生成该物质 ($\nu_B=+1$) 反应的标准摩尔熵。

2.11

$$\begin{array}{lll} \Delta H^\theta: & -40 \text{ kJ} & -40 \text{ kJ} \\ \Delta G^\theta: & -38 \text{ kJ} & -38 \text{ kJ} \\ \Delta S^\theta: & -6.7 \text{ J} \cdot \text{K}^{-1} & -6.7 \text{ J} \cdot \text{K}^{-1} \end{array}$$

2.12

$$\begin{aligned} S_n(\text{白锡}) &= S_n(\text{灰锡}) \\ \Delta_r H_m^\theta(298\text{K}) &= \Delta_f H_m^\theta(\text{灰锡}) + (-1) \times \Delta_f H_m^\theta(\text{白锡}) = -2.1 \text{ kJ} \cdot \text{mol}^{-1} < 0 \\ \Delta_r S_m^\theta(298\text{K}) &= S_m^\theta(\text{灰锡}) - S_m^\theta(\text{白锡}) = -7.41 \text{ J} \cdot \text{mol}^{-1} \cdot \text{K}^{-1} < 0 \\ \therefore \text{反应在标准状态, 低温自发。在转变温度, } \Delta_r G_m^\theta(T) &= 0 \\ \therefore \Delta_r G_m^\theta(T) &\approx \Delta_r H_m^\theta(298\text{K}) - T \Delta_r S_m^\theta(298\text{K}) \\ T &\approx [\Delta_r H_m^\theta(298\text{K}) / \Delta_r S_m^\theta(298\text{K})] \approx 283\text{K} \end{aligned}$$

2.13

$$\begin{aligned} 2\text{Fe}_2\text{O}_3(\text{s}) + 3\text{C}(\text{s}) &= 4\text{Fe}(\text{s}) + 3\text{CO}_2(\text{g}) \\ \Delta_r H_m^\theta(298\text{K}) &= 3\Delta_f H_m^\theta(\text{CO}_2, \text{g}) + (-2) \times \Delta_f H_m^\theta(\text{Fe}_2\text{O}_3, \text{s}) = 467.87 \text{ kJ} \cdot \text{mol}^{-1} \\ \Delta_r S_m^\theta(298\text{K}) &= 4 \times S_m^\theta(\text{Fe}, \text{s}) + 3 S_m^\theta(\text{CO}_2, \text{g}) + (-2) \times S_m^\theta(\text{Fe}_2\text{O}_3, \text{s}) + \\ &\quad (-3) \times S_m^\theta(\text{C}, \text{s}) = 557.98 \text{ J} \cdot \text{mol}^{-1} \cdot \text{K}^{-1} \end{aligned}$$

反应在标准状态下自发进行:

$$\begin{aligned} \Delta_r G_m^\theta(T) &\approx \Delta_r H_m^\theta(298\text{K}) - T \Delta_r S_m^\theta(298\text{K}) \\ T &> [\Delta_r H_m^\theta(298\text{K}) / \Delta_r S_m^\theta(298\text{K})], \quad \text{即 } T > 839\text{K} \end{aligned}$$

故以木炭为燃料时, 因最高温度低于 839K, 反应不可能自发。

$$\begin{aligned} 2\text{CuO}(\text{s}) + \text{C}(\text{s}) &= 2\text{Cu}(\text{s}) + \text{CO}_2(\text{g}) \\ \Delta_r H_m^\theta(298\text{K}) &= -79.51 \text{ kJ} \cdot \text{mol}^{-1} > 0 \\ \Delta_r S_m^\theta(298\text{K}) &= 189 \text{ J} \cdot \text{mol}^{-1} \cdot \text{K}^{-1} > 0 \end{aligned}$$

$\therefore$ 反应在标准状态, 任意温度均自发

2.14 (略)

$$\begin{aligned} 2.15 \quad 2\text{AsH}_3(\text{g}) &= 2\text{As}(\text{s}) + 3\text{H}_2(\text{g}) \\ \Delta_r H_m^\theta(298\text{K}) &= -132.8 \text{ kJ} \cdot \text{mol}^{-1} < 0 \\ \Delta_r S_m^\theta(298\text{K}) &= 15.71 \text{ J} \cdot \text{mol}^{-1} \cdot \text{K}^{-1} < 0 \end{aligned}$$

$\therefore$ 标准状态, 任意温度下 AsH_3 的分解反应均自发。加热的目的是加快反应速率。

第三章 化学平衡原理

3.1 (1) 正确, (2) 错误, (3) 错误。

3.2 $K^\ominus = 40$

3.3

$$K^\ominus = \frac{\{c(\text{Fe}^{2+})/c^\ominus\}\{p(\text{H}_2)/p^\ominus\}}{\{c(\text{H}^+)/c^\ominus\}^2}$$

3.4 $4 \times (3) - 2 \times (1) - 2 \times (2)$ 得所求反应式

$$\text{故: } \Delta_r G_m^\ominus = 4 \Delta_r G_m^\ominus(3) - 2 \Delta_r G_m^\ominus(1) - 2 \Delta_r G_m^\ominus(2)$$

$$K^\ominus = \frac{\{K^\ominus(3)\}^4}{\{K^\ominus(1)\}^2 \{K^\ominus(2)\}^2}$$

$$p = \frac{nRT}{V} = \frac{0.20 \text{ mol} \times 8.31 \text{ kPa} \cdot \text{L} \cdot \text{mol}^{-1} \cdot \text{K}^{-1} \times 698 \text{ K}}{10 \text{ L}} = 116 \text{ kPa}$$

$$n(\text{I}_2) = n(\text{H}_2) = c(\text{I}_2)V = 0.0021 \text{ mol} \cdot \text{L}^{-1} \times 10 \text{ L} = 0.021 \text{ mol}$$

$$p(\text{I}_2) = p(\text{H}_2) = x(\text{I}_2) \cdot p = \frac{0.021 \text{ mol} \times 116 \text{ kPa}}{0.20 \text{ mol}} = 12.2 \text{ kPa}$$

$$p(\text{HI}) = p - p(\text{I}_2) - p(\text{H}_2) = 91.6 \text{ kPa}$$

$$K^\ominus = \frac{\{p(\text{HI})/p^\ominus\}^2}{\{p(\text{H}_2)/p^\ominus\}\{p(\text{I}_2)/p^\ominus\}} = 56$$

$$K^\ominus = \frac{c(\text{O}_2)/c^\ominus}{p(\text{O}_2)/p^\ominus}$$

$$K^\ominus(293 \text{ K}) = \frac{1.38 \times 10^{-3} \text{ mol} \cdot \text{L}^{-1} / 1 \text{ mol} \cdot \text{L}^{-1}}{101 \text{ kPa} / 100 \text{ kPa}} = 1.37 \times 10^{-3}$$

3.5 $pV = nRT$

故对于反应 $\text{H}_2(\text{g}) + \text{I}_2(\text{g}) = 2\text{HI}(\text{g})$

3.6

(1) $\text{O}_2(\text{g}) = \text{O}_2(\text{aq})$

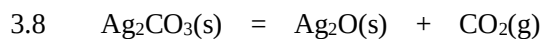
$P(\text{O}_2) = 21.0 \text{ kPa}$ 时:

$$K^{\ominus}(293\text{K}) = \frac{c(\text{O}_2)/c^{\ominus}}{0.21} = 1.37 \times 10^{-3}$$

故: $c(\text{O}_2, \text{aq}) = 2.88 \times 10^{-4} \text{ mol} \cdot \text{L}^{-1}$

(2) $K^{\ominus} = 85.5 / 1.37 \times 10^{-3} = 6.24 \times 10^4$

3.7 $Q = 4$ 逆向自发 $Q = 0.1$ 逆向自发



$$\Delta_r G_m^{\ominus}(383\text{K}) = 14.8 \text{ kJ} \cdot \text{mol}^{-1}$$

$$\ln K^{\ominus}(383\text{K}) = -\Delta_r G_m^{\ominus}(383\text{K})/RT = -4.65$$

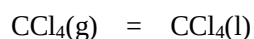
$$K^{\ominus}(383\text{K}) = 9.56 \times 10^{-3}$$

$$K^{\ominus} = p(\text{CO}_2)/p^{\ominus}$$

为防止反应正向自发, 应保证 $Q > K^{\ominus}$

故: $p(\text{CO}_2) > 9.56 \times 10^{-1} \text{ kPa}$

3.9	$\text{CCl}_4(\text{g})$	$\text{CCl}_4(\text{l})$
$\Delta_f H_m^{\ominus}/\text{kJ} \cdot \text{mol}^{-1}$	-102.93	-135.4
$S_m^{\ominus}/\text{J} \cdot \text{K}^{-1} \cdot \text{mol}^{-1}$	309.74	216.4



$$\Delta_r H_m^{\ominus} = 32.47 \text{ kJ} \cdot \text{mol}^{-1}$$

$$\Delta_r S_m^{\ominus} = 0.09334 \text{ kJ} \cdot \text{mol}^{-1} \cdot \text{K}^{-1}$$

所以 CCl_4 的正常沸点为 $T_1 = 348\text{K}$

根据克拉贝龙——克劳休斯方程计算, 可得在 20kPa 时, 其沸点为 304K 。

第四章 化学反应速率

4.1 不能

4.2 A 的物理意义为 PZ_0 , f 的物理意义为 相对平均动能超过 E_c 的 A、B 分子间的碰撞频率与 A、B 分子间碰撞频率的比值。4.3 Z_0 基本不变, P 不变, f 大, k 大, v 大。

4.4 直线

$$4.5 \quad -\frac{dc(\text{H}_2)}{dt} = 6.0 \text{ mol} \cdot \text{L}^{-1} \cdot \text{s}^{-1}, \quad -\frac{dc(\text{N}_2)}{dt} = 4.0 \text{ mol} \cdot \text{L}^{-1} \cdot \text{s}^{-1},$$

4.6 略

4.7 解:

速率方程为 $v = kc^m(\text{NOCl})$

$$(1) \quad v_1 = k(0.200 \text{ mol} \cdot \text{L}^{-1})^m = \frac{4.8 \times 10^{-5}}{2} \text{ mol} \cdot \text{L}^{-1} \cdot \text{s}^{-1} \quad (2) \quad v_2 = k(0.400 \text{ mol} \cdot \text{L}^{-1})^m = \frac{1.92 \times 10^{-4}}{2} \text{ mol} \cdot \text{L}^{-1} \cdot \text{s}^{-1}$$

$$(1)/(2) \text{ 得: } \left(\frac{0.200 \text{ mol} \cdot \text{L}^{-1}}{0.400 \text{ mol} \cdot \text{L}^{-1}} \right)^m = \frac{4.8 \times 10^{-5} \text{ mol} \cdot \text{L}^{-1} \cdot \text{s}^{-1}}{1.92 \times 10^{-4} \text{ mol} \cdot \text{L}^{-1} \cdot \text{s}^{-1}} \quad m \approx 2$$

$$k = \frac{v}{c^2(\text{NOCl})} = \frac{4.8 \times 10^{-5} \text{ mol} \cdot \text{L}^{-1} \cdot \text{s}^{-1} / 2}{(0.200 \text{ mol} \cdot \text{L}^{-1})^2} = 6.0 \times 10^{-4} \text{ mol}^{-1} \cdot \text{L} \cdot \text{s}^{-1}$$

$$v_3 = 6.0 \times 10^{-4} \text{ mol}^{-1} \cdot \text{L} \cdot \text{s}^{-1} \times (0.500 \text{ mol} \cdot \text{L}^{-1})^2 = 1.5 \times 10^{-4} \text{ mol} \cdot \text{L}^{-1} \cdot \text{s}^{-1}$$

4.8

(1) 反应的速率方程为 $v = kc^m(\text{NO})c^n(\text{O}_2)$

(2) 将表中数据代入上式得

$$(a) \quad v_1 = k(0.10 \text{ mol} \cdot \text{L}^{-1})^m (0.10 \text{ mol} \cdot \text{L}^{-1})^n = \frac{3.0 \times 10^{-2}}{2} \text{ mol} \cdot \text{L}^{-1} \cdot \text{s}^{-1} \quad (b) \quad v_2 = k(0.10 \text{ mol} \cdot \text{L}^{-1})^m (0.20 \text{ mol} \cdot \text{L}^{-1})^n = \frac{6.0 \times 10^{-2}}{2} \text{ mol} \cdot \text{L}^{-1} \cdot \text{s}^{-1}$$

$$(c) \quad v_3 = k(0.30 \text{ mol} \cdot \text{L}^{-1})^m (0.20 \text{ mol} \cdot \text{L}^{-1})^n = \frac{0.54}{2} \text{ mol} \cdot \text{L}^{-1} \cdot \text{s}^{-1}$$

$$(a)/(b) \text{ 得: } \left(\frac{1}{2} \right)^n = \frac{1}{2} \quad n = 1$$

$$(b)/(c) \text{ 得: } \left(\frac{1}{3} \right)^n = \frac{1}{9} \quad m = 2(b)/(c) \text{ 得: } \left(\frac{1}{3} \right)^n = \frac{1}{9} \quad m = 2$$

$$k = \frac{v}{c^2(\text{NO})c(\text{O}_2)} = \frac{3.0 \times 10^{-2} \text{ mol} \cdot \text{L}^{-1} \cdot \text{s}^{-1} / 2}{(0.10 \text{ mol} \cdot \text{L}^{-1})^2 (0.10 \text{ mol} \cdot \text{L}^{-1})} = 15 \text{ mol}^{-2} \cdot \text{L}^2 \cdot \text{s}^{-1}$$

$$(3) v_3 = 15 \text{ mol}^{-2} \cdot \text{L}^2 \cdot \text{s}^{-1} (0.15 \text{ mol} \cdot \text{L}^{-1})^2 (0.15 \text{ mol} \cdot \text{L}^{-1}) = 5.0 \times 10^{-2} \text{ mol} \cdot \text{L}^{-1} \cdot \text{s}^{-1}$$

4.9 解:

$$\frac{k_{t+n \times 10}}{k_t} = r^n \quad \frac{k_{30}}{k_{20}} = r \quad \frac{3.5 \times 10^{-5} \text{ mol}^{-1} \cdot \text{L} \cdot \text{s}^{-1}}{1.3 \times 10^{-5} \text{ mol}^{-1} \cdot \text{L} \cdot \text{s}^{-1}} = r \quad r = 2.7$$

$$\frac{k_{20+3 \times 10}}{k_{20}} = r^3 \quad k_{50} = k_{20} \times r^3 = 1.3 \times 10^{-5} \text{ s}^{-1} \times (2.7)^3 = 2.6 \times 10^{-6} \text{ s}^{-1}$$

4.10 解:

$$\ln \frac{k_2}{k_1} = \frac{E_a}{R} \left(\frac{T_2 - T_1}{T_2 T_1} \right)$$

$$\ln \frac{2.0 \times 10^{-5} \text{ s}^{-1}}{k_1} = \frac{226 \times 10^3 \text{ J} \cdot \text{mol}^{-1}}{8.314 \text{ J} \cdot \text{mol}^{-1} \cdot \text{K}} \left(\frac{650 \text{ K} - 600 \text{ K}}{650 \text{ K} \times 600 \text{ K}} \right) \quad k_1 = 6.1 \times 10^{-7} \text{ s}^{-1}$$

第五章 原子结构

5.1 原子发射光谱, 电子衍射

5.2 确定, 动量, 不确定, 轨道

5.3 (2), (4)正确

5.4 波函数, 电子的几率密度

5.5 (1)

5.6

	n	l	m	m_s
1	2	0	0	+1/2
2	2	0	0	- 1/2
3	2	1	0	+1/2
4	2	1	+1	+1/2
5	2	1	-1	+1/2

5.7 (3)

5.8 $\text{He}^+ \quad E(3s) = E(3p) = E(3d) < E(4s)$

$\text{K} \quad E(3s) < E(3p) < E(4s) < E(3d)$

$\text{Mn} \quad E(3s) < E(3p) < E(3d) < E(4s)$

5.9 (略)

5.10 4s, 3d, 4p

5.11 能级组, 外层电子结构, 主族元素基态原子内层轨道,或全满,或全空(稳定构型)

5.12 一,二,三主族元素,镧系元素, 第六周期镧后元素

5.13 $\text{He} > \text{H}, \text{Ba} > \text{Sr}, \text{Ca} > \text{Sc}, \text{Cu} > \text{Ni}, \text{La} > \text{Y}, \text{Zr} > \text{Ti}, \text{Zr} > \text{Hf}$

5.14 (3), (2)

5.15 Mg 失去 2 个最外层 s 电子后成稳定的电子构型, 故 I_3 明显大于 I_2 , 常见氧化数为+2;
 Al 失去 3 个最外层电子后成稳定的电子构型, 故 I_4 明显大于 I_3 , 常见氧化数为+3。 Mg 元素
 第一、第二电离能之和小于 Al 元素第一、第二、第三电离能之和, 所以气态 Mg 原子的金属性
 强于 Al 。

第六章 化学键与分子结构

6.1 (1) 错, (2) 错, (3) 错, (4) 错, (5) 对, (6) 错

6.2 离子, $\text{BeO} > \text{MgO} > \text{CaO} > \text{SrO} > \text{BaO}$,
 $\text{BaCO}_3 > \text{SrCO}_3 > \text{CaCO}_3 > \text{MgCO}_3 > \text{BeCO}_3$

6.3 Fe^{2+} : $[\text{Ar}]3\text{d}^6$, 9~17,
 Fe^{3+} : $[\text{Ar}]3\text{d}^5$, 9~17,
 Pb^{2+} : $[\text{Xe}]5\text{d}^{10}6\text{s}^2$ 18+2,
 Sn^{4+} : $[\text{Kr}]4\text{d}^{10}$, 18,
 Al^{3+} : $[\text{He}]2\text{s}^22\text{p}^6$, 8,
 S^{2-} : $[\text{Ne}]3\text{s}^23\text{p}^6$, 8,
 Hg^{2+} : $[\text{Xe}]5\text{d}^{10}$, 18.

6.4 OF_2 : 非等性 sp^3 杂化, V 型, 极性;
 NF_3 : 非等性 sp^3 杂化, 三角锥, 极性;
 BH_3 : 等性 sp^2 杂化, 正三角型, 非极性;
 SiCl_4 : 等性 sp^3 杂化, 正四面体, 非极性;
 NH_3 : 非等性 sp^3 杂化, 三角锥, 极性;
 HCN : 等性 sp 杂化, 直线型, 极性;
 PCl_3 : 非等性 sp^3 杂化, 三角锥, 极性;
 PCl_5 : sp^3d 杂化, 三角双锥, 非极性;
 CS_2 : sp 杂化, 直线型, 非极性;
 SnCl_2 : 非等性 sp^3 杂化, V 型, 极性。

6.5 C_2H_6 : sp^3 ; C_2H_4 : sp^2 ;
 CH_3CCH : sp^3 sp sp ;
 $\text{CH}_3\text{CH}_2\text{OH}$: sp^3 ; H_2CO : sp^2 ; COCl_2 : sp^2 .

6.6 正丁烷: 否;
 1, 3-丁二烯: 否;
 2-丁炔: 是

6.7 HF HCl HBr HI ; HF HCl HBr HI ; HI HBr HCl HF ; HF HI HBr HCl

6.8 (1) ~ c; (2) ~ d; (3) ~ a; (4) ~ b

- 6.9 (1) Fe^{3+} 电荷高、半径小，极化力大于 Fe^{2+} ;
 (2) Zn^{2+} 18 电子构型，极化力、变形性大于 Ca^{2+} ;
 (3) Cd^{2+} 18 电子够型，极化力、变形性大于 Mn^{2+} 。

6.10 ClO^- , ClO_2^- , ClO_3^- , ClO_4^- 。

6.11 HClO 酸性强于 HBrO 。成酸元素原子电负性大，含氧酸酸性强。

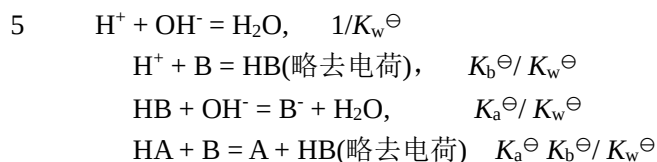
第七章 酸碱反应

1 (略)

2 H_3O^+ , OH^-

3 (2)

4 (2)



- 6 (1) $K^\ominus(\text{正}) = 1.9 \times 10^5 > K^\ominus(\text{逆})$, 易于正向反应
 (2) $K^\ominus(\text{正}) = 2 \times 10^{-11} < K^\ominus(\text{逆})$, 易于逆向反应
 (3) $K^\ominus(\text{正}) = 1.4 \times 10^2 > K^\ominus(\text{逆})$, 易于正向反应
 (4) $K^\ominus(\text{正}) = 6.2 \times 10^{-8} < K^\ominus(\text{逆})$, 易于逆向反应

7 6.4×10^{-5} , 1.7×10^{-13} , 小于

7.8

$$\frac{c/c^\ominus}{K_b^\ominus} > 10^{2.81}$$

$$c(\text{OH}^-)/c^\ominus = \sqrt{K_b^\ominus(c/c^\ominus)} = 3.74 \times 10^{-4}$$

$$\text{pH} = 10.57$$

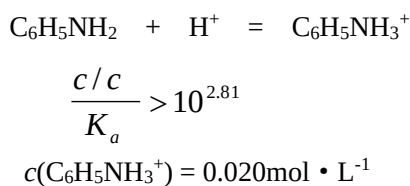
7.9

$$c(\text{H}^+)/c^\ominus = 8.5 \times 10^{-5}$$

$$c^{\text{eq}}(\text{HNO}_2)/c^\ominus = \{c(\text{H}^+)/c^\ominus\}^2 / K_a^\ominus = 1.6 \times 10^{-5}$$

$$\therefore c_0(\text{HNO}_2) = 1.6 \times 10^{-5} \text{ mol} \cdot \text{L}^{-1} + 8.5 \times 10^{-5} \text{ mol} \cdot \text{L}^{-1} = 1.0 \times 10^{-4} \text{ mol} \cdot \text{L}^{-1}$$

7.10



$$K_a^\ominus(\text{C}_6\text{H}_5\text{NH}_3^+) = K_w^\ominus / K_b^\ominus(\text{C}_6\text{H}_5\text{NH}_2) = 2.2 \times 10^{-5}$$

$$\text{应用最简式计算溶液酸度: } c(\text{H}^+) = 6.6 \times 10^{-4} \text{ mol/L} \quad \text{pH} = 3.2$$

7.11

$$K_{b1}^\ominus / K_{b2}^\ominus > 10^{1.6}$$

$$\frac{c/c^\ominus}{K_{b1}^\ominus} = 7 \times 10^4 > 10^{2.81}$$

$$c(\text{OH}^-)/c^\ominus = \sqrt{K_{b1}^\ominus \times c/c^\ominus} = \sqrt{7.0 \times 10^{-7} \times 0.050} = 1.9 \times 10^{-4}$$

$$\text{pH} = 10.27$$

所以可忽略第二步离解，按最简式计算：

$$c(\text{C}_{10}\text{H}_{14}\text{N}_2) = 0.050 \text{ mol} \cdot \text{L}^{-1} - 1.9 \times 10^{-4} \text{ mol} \cdot \text{L}^{-1} = 0.050 \text{ mol} \cdot \text{L}^{-1}$$

$$c(\text{C}_{10}\text{H}_{14}\text{N}_2\text{H}^+) = c(\text{OH}^-) = 1.9 \times 10^{-4} \text{ mol} \cdot \text{L}^{-1}$$

$$c(\text{C}_{10}\text{H}_{14}\text{N}_2\text{H}_2^{2+}) = c^\ominus K_{b2}^\ominus = 1.4 \times 10^{-11} \text{ mol} \cdot \text{L}^{-1}$$

7.12

$$K_{b1}^\ominus(\text{S}^{2-}) = K_w^\ominus / K_{a2}^\ominus(\text{H}_2\text{S}) = 1.4$$

$$K_{b2}^\ominus(\text{S}^{2-}) = K_w^\ominus / K_{a1}^\ominus(\text{H}_2\text{S}) = 7.7 \times 10^{-8}$$

经判断，可忽略第二步离解，依近似式计算

$$\begin{array}{ccccccc} \text{S}^{2-} & + & \text{H}_2\text{O} & = & \text{HS}^- & + & \text{OH}^- \\ \text{C}^{\text{eq}}/\text{mol} \cdot \text{L}^{-1} & & 0.10-x & & x & & x \end{array}$$

$$K_{b1}^\ominus = \frac{x^2}{0.10-x} = 1.4$$

$$x = 0.094$$

$$\text{即: } c(\text{OH}^-) = 0.094 \text{ mol} \cdot \text{L}^{-1}$$

$$\text{pH} = 12.97$$

$$c(\text{S}^{2-}) = 0.10 \text{ mol} \cdot \text{L}^{-1} - 0.094 \text{ mol} \cdot \text{L}^{-1} = 0.01 \text{ mol} \cdot \text{L}^{-1}$$

$$c(\text{HS}^-) = c(\text{OH}^-) = 0.094 \text{ mol} \cdot \text{L}^{-1}$$

$$c(\text{H}_2\text{S}) = c^\ominus K_{b2}^\ominus = 7.7 \times 10^{-8} \text{ mol} \cdot \text{L}^{-1}$$

7.13

$$\text{pH} = 1.30$$

$$c(\text{S}^{2-})/c^\ominus = \frac{K_{a1}^\ominus K_{a2}^\ominus \{c(\text{H}_2\text{S})/c^\ominus\}}{\{c(\text{H}^+)/c^\ominus\}^2} = \frac{9.2 \times 10^{-22} \times 0.10}{0.050^2} = 3.7 \times 10^{-20}$$

$$\text{故: } c(\text{S}^{2-}) = 3.7 \times 10^{-20} \text{ mol} \cdot \text{L}^{-1}$$

7.14

$$\text{pH} = \text{p}K_{a1}^\ominus - \lg \frac{c(\text{H}_3\text{Cit})}{c(\text{H}_2\text{Cit}^-)}$$

$$\lg \frac{c(\text{H}_3\text{Cit})}{c(\text{H}_2\text{Cit}^-)} = 3.15 - 1.92 = 1.23$$

$$\text{pH} = \text{p}K_{a2}^\ominus - \lg \frac{c(\text{H}_2\text{Cit}^-)}{c(\text{HCit}^{2-})} = 1.7$$

$$\frac{c(\text{H}_2\text{Cit}^-)}{c(\text{HCit}^{2-})} = 10$$

$$\begin{aligned} \text{pH} &= \text{p}K_{a3}^{\ominus} - \lg \frac{c(\text{HCit}^{2-})}{c(\text{Cit}^{3-})} \\ \lg \frac{c(\text{HCit}^{2-})}{c(\text{Cit}^{3-})} &= 6.39 - 1.92 = 4.47 \\ \therefore \frac{c(\text{HCit}^{2-})}{c(\text{Cit}^{3-})} &= 3.0 \times 10^4 \\ \frac{c(\text{H}_2\text{Cit}^-)}{c(\text{Cit}^{3-})} &= 3.0 \times 10^4 \times 7.0 \times 10^2 = 2.1 \times 10^7 \\ \frac{c(\text{H}_3\text{Cit})}{c(\text{Cit}^{3-})} &= 3.0 \times 10^4 \times 7.0 \times 10^2 \times 17 = 3.6 \times 10^8 \end{aligned}$$

7.15 $\text{pOH} = 5.00$

$$\begin{aligned} \text{pOH} &= \text{p}K_b^{\ominus}(\text{NH}_3) - \lg \frac{c(\text{NH}_3)}{c(\text{NH}_4^+)} \\ \lg \frac{c(\text{NH}_3)}{c(\text{NH}_4^+)} &= 4.75 - 5.00 = -0.25 \quad \therefore \frac{c(\text{NH}_3)}{c(\text{NH}_4^+)} = 0.56 \\ c(\text{NH}_4^+) &= 0.10 \text{mol/L} / 0.56 = 0.18 \text{mol/L} \\ M\{(\text{NH}_4)_2\text{SO}_4\} &= 0.5c(\text{NH}_4^+)V(\text{NH}_4^+)M\{(\text{NH}_4)_2\text{SO}_4\} = 11.9 \text{g} \end{aligned}$$

7.16

$$\begin{aligned} \text{pH} &= \frac{1}{2}(\text{p}K_{a1}^{\ominus} + \text{p}K_{a2}^{\ominus}) = 3.7 \\ \text{pH} &= \text{p}K_{a1}^{\ominus} - \lg \frac{c(\text{H}_2\text{A})}{c(\text{HA}^-)} \\ \therefore \lg \frac{c(\text{H}_2\text{A})}{c(\text{HA}^-)} &= \text{p}K_{a1}^{\ominus} - \text{pH} = \frac{1}{2}(\text{p}K_{a1}^{\ominus} - \text{p}K_{a2}^{\ominus}) = -0.7 \quad \frac{c(\text{H}_2\text{A})}{c(\text{HA}^-)} = 0.20 \\ \text{pH} &= \text{p}K_{a2}^{\ominus} + \lg \frac{c(\text{A}^{2-})}{c(\text{HA}^-)} \\ \therefore \lg \frac{c(\text{A}^{2-})}{c(\text{HA}^-)} &= \text{pH} - \text{p}K_{a2}^{\ominus} = \frac{1}{2}(\text{p}K_{a1}^{\ominus} - \text{p}K_{a2}^{\ominus}) = -0.7 \quad \frac{c(\text{A}^{2-})}{c(\text{HA}^-)} = 0.20 \end{aligned}$$

因系统中含有 2 缓冲对，且酸与碱的浓度比均在缓冲范围内，所以此溶液具有酸碱缓冲能力。

若两级酸常数相差较大，则酸碱浓度比将超出缓冲范围，失去缓冲性质。

7.17 甘氨酸：不移动；谷氨酸：向正极运动；赖氨酸：向负极运动

第八章 沉淀—溶解反应

8.1

$$K_{sp}^{\ominus}(\text{AgBr}) = \{c(\text{Ag}^+)/c^{\ominus}\} \{c(\text{Br}^-)/c^{\ominus}\}$$

$$K_{sp}^{\ominus}(\text{Ag}_2\text{S}) = \{c(\text{Ag}^+)/c^{\ominus}\}^2 \{c(\text{S}^{2-})/c^{\ominus}\}$$

$$K_{sp}^{\ominus}(\text{Ca}_3(\text{PO}_4)_2) = \{c(\text{Ca}^{2+})/c^{\ominus}\}^3 \{c(\text{PO}_4^{3-})/c^{\ominus}\}$$

$$K_{sp}^{\ominus}(\text{MgNH}_4\text{AsO}_4) = \{c(\text{Ca}^{2+})/c^{\ominus}\} \{c(\text{NH}_4^+)/c^{\ominus}\} \{c(\text{AsO}_4^{3-})/c^{\ominus}\}$$

8.2

(1)

$$s = c^{\ominus} \sqrt[3]{\frac{K_{sp}^{\ominus}}{4}} = 1.12 \times 10^{-4} \text{ mol/L}$$

(2)

$$s = c(\text{Mg}^{2+}) = \frac{c^{\ominus} K_{sp}^{\ominus}}{\{c(\text{OH}^-)/c^{\ominus}\}^2} = 5.6 \times 10^{-8} \text{ mol/L}$$

(3)

$$s = \frac{1}{2} c(\text{OH}^-) = \frac{c^{\ominus}}{2} \sqrt{\frac{K_{sp}^{\ominus}}{c(\text{Mg}^{2+})/c^{\ominus}}} = 1.2 \times 10^{-5} \text{ mol/L}$$

8.3 PbCO_3 计算结果偏低

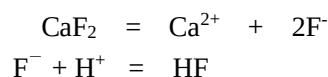
8.4 因氢氧化铁在水中溶解度极小, 溶液 pH 约等于 7.0

$$s = c(\text{Fe}^{3+}) = c^{\ominus} \times \frac{K_{sp}^{\ominus}}{\{c(\text{OH}^-)/c^{\ominus}\}^3} = \frac{2.64 \times 10^{-39} \text{ mol/L}}{(1.0 \times 10^{-7})^3} = 2.6 \times 10^{-18} \text{ mol/L}$$

8.5

(2)

8.6



$$\therefore \text{pH} = \text{p}K_a^{\ominus}(\text{HF}) - \lg \frac{c(\text{HF})}{c(\text{F}^-)} \quad \therefore \frac{c(\text{HF})}{c(\text{F}^-)} = 0.028$$

根据: $2c(\text{Ca}^{2+}) = c(\text{HF}) + c(\text{F}^-) = 2s$, 得: $c(\text{F}^-) = 2s / 1.028$

$$K_{sp}^{\ominus}(\text{CaF}_2) = (s/c^{\ominus})(2s/1.028c^{\ominus})^2 \quad S = 3.38 \times 10^{-4} \text{ mol/L}$$

8.7 CaF_2 、 CaCO_3 和 MnS , KClO_4

8.8 此系统为多重平衡系统, 所以:

$$\frac{K_{sp}^{\ominus}(\text{CaCO}_3)}{K_{sp}^{\ominus}(\text{CaF}_2)} = \frac{c(\text{CO}_3^{2-})/c^{\ominus}}{\{c(\text{F}^-)/c^{\ominus}\}^2}$$

$$c(\text{CO}_3^{2-}) = \frac{4.96 \times 10^{-9}}{1.46 \times 10^{-10}} \times (2.0 \times 10^{-4})^2 \text{ mol/L} = 1.36 \times 10^{-6} \text{ mol/L}$$

$$8.9 \quad c(\text{NH}_3) = 0.050 \text{ mol/L}$$

$$\therefore c(\text{OH}^-)/c^\ominus = \sqrt{K_b^\ominus(c/c^\ominus)} = \sqrt{1.76 \times 10^{-5} \times 0.050} = 9.5 \times 10^{-4}$$

$$\{c(\text{Mg}^{2+})/c^\ominus\} \{c(\text{OH}^-)/c^\ominus\}^2 = 0.25 \times (9.5 \times 10^{-4})^2 = 2.3 \times 10^{-7} > K_{sp}^\ominus\{\text{Mg}(\text{OH})_2\}$$

故有氢氧化镁沉淀生成。

为防止沉淀生成, $c(\text{OH}^-)/c^\ominus$ 的最大值为:

$$c(\text{OH}^-)/c^\ominus = \sqrt{\frac{K_{sp}^\ominus}{c(\text{Mg}^{2+})/c^\ominus}} = 4.74 \times 10^{-6}$$

$$\text{pOH} = 5.32$$

$$\therefore \text{pOH} = \text{p}K_b^\ominus(\text{NH}_3) - \lg \frac{c(\text{NH}_3)}{c(\text{NH}_4^+)}$$

$$\lg \frac{c(\text{NH}_3)}{c(\text{NH}_4^+)} = 4.75 - 5.32 = -0.57$$

$$\therefore c(\text{NH}_4^+) = \frac{0.050 \text{ mol/L}}{0.27} = 0.19 \text{ mol/L}$$

$$m(\text{NH}_4\text{Cl}) = c(\text{NH}_4^+) \cdot V \cdot M(\text{NH}_4\text{Cl}) = 0.41 \text{ g}$$

8.10

故不被沉淀的离子 M^{2+} 的硫化物的溶度积最小为:

$$K_{sp}^\ominus = \{c(\text{M}^{2+})/c^\ominus\} \{c(\text{S}^{2-})/c^\ominus\} = 0.01 \times 1.0 \times 10^{-21} = 1 \times 10^{-23} \quad \text{所以, } \text{Mn}^{2+}, \text{Ni}^{2+} \text{ 不被沉淀。}$$

离子被完全定性沉淀时, $c(\text{M}^{2+}) \leq 10^{-5} \text{ mol/L}$, 故可被沉淀完全的硫化物溶度积的最大值为:

$$c(\text{S}^{2-})/c^\ominus = \frac{K_{a1}^\ominus K_{a2}^\ominus \{c(\text{H}_2\text{S})/c^\ominus\}}{\{c(\text{H}^+)/c^\ominus\}^2} = \frac{9.2 \times 10^{-12} \times 0.10}{0.30^2} = 1.0 \times 10^{-21}$$

$$K_{sp}^\ominus = 10^{-5} \times 1 \times 10^{-21} = 10^{-26}$$

所以可被沉淀完全的离子为: $\text{Pb}^{2+}, \text{Cd}^{2+}, \text{Hg}^{2+}, \text{Cu}^{2+}$ 。

8.11 欲使 $\text{Fe}(\text{OH})_3$ 沉淀完全:

$$c(\text{OH}^-)/c^\ominus > \sqrt[3]{\frac{K_{sp}^\ominus\{\text{Fe}(\text{OH})_3\}}{10^{-5}}}$$

$$c(\text{OH}^-)/c^\ominus > 6.4 \times 10^{-12}$$

$$\text{pH} > 2.78$$

欲使 $\text{Fe}(\text{OH})_2$ 不发生沉淀:

所以应控制 pH 约在 2.8~6.5 范围。

8.13 CuCO_3 (计算得 CuCO_3 的溶解度为 $1.2 \times 10^{-5} \text{mol/L}$, 即 CuCO_3 饱和水溶液的体积浓度为 0.76mg/L)

$$\{c(\text{Ca})/c^\ominus\}\{c(\text{CO}_3^{2-})/c^\ominus\}=5.61\times 10^{-12}<K_{\text{sp}}^\ominus(\text{CaCO}_3)$$

8.15 (有关数据: $K_{sp}^{\ominus}(\text{CaCO}_3)=4.96 \times 10^{-9}$, $K_{sp}^{\ominus}(\text{ZnCO}_3)=1.19 \times 10^{-10}$,

在 $c(\text{CO}_3^{2-}) = 0.10\text{mol/L}$ 的碳酸钠水溶液中:

对于两价离子 M^{2+} 的氢氧化物:

对于两价离子的碳酸盐:

所以生成的沉淀是:

$$\text{CaCO}_3, \text{Mg}_2(\text{OH})_2\text{CO}_3, \text{Zn}_2(\text{OH})_2\text{CO}_3, \text{Ni}_2(\text{OH})_2\text{CO}_3$$

对于三价离子 Fe^{3+} :

所以生成 $\text{Fe}(\text{OH})_3$

16 (4)

17 溶度积, 离子浓度, 沉淀类型。

得: $s = 8.4 \times 10^{-3} \text{ mol/L}$

$$s = c(\text{SO}_4^{2-})M(\text{SO}_4^{2-}) = 8.4 \times 10^{-3} \text{ mol/L} \times 9.6 \times 10^4 \text{ mg/mol}$$
$$= 806 \text{ mg/L}$$

所以不可饮用。

第九章 氧化还原反应

9.3 还原, 氧化

9.4 不变, 不变, 变为 $\{K^\ominus\}^n$

9.5 (1)

9.6 (2)

9.7 (4)

9.8 (3)

9.9 (1) $2\text{Fe}^{3+} + \text{Sn}^{2+} = 2\text{Fe}^{2+} + \text{Sn}^{4+}$

(2) $2\text{Fe}^{3+} + \text{Cu} = 2\text{Fe}^{2+} + \text{Cu}^{2+}$

(3) $2\text{MnO}_4^- + 10\text{Cl}^- + 16\text{H}^+ = 2\text{Mn}^{2+} + 5\text{Cl}_2 + 8\text{H}_2\text{O}$

(4) $\text{H}_2\text{O}_2 + 2\text{Fe}^{2+} + 2\text{H}^+ = 2\text{Fe}^{3+} + 2\text{H}_2\text{O}$

(5) $\text{PbO}_2 + 2\text{Cl}^- + 4\text{H}_2\text{O} = \text{Pb}^{2+} + \text{Cl}_2 + 2\text{H}_2\text{O}$

(6) $\text{Hg}_2\text{Cl}_2 + \text{Sn}^{2+} = 2\text{Hg} + \text{Sn}^{4+} + 2\text{Cl}^-$

(7) $2\text{MnO}_4^- + 3\text{Mn}^{2+} + 2\text{H}_2\text{O} = 5\text{MnO}_2 + 4\text{H}^+$

9.10 (1)

$\text{Zn} | \text{Zn}^{2+}(0.020\text{mol/L}) || \text{Ni}^{2+}(0.080\text{mol/L}) | \text{Ni}$

$$\varphi(\text{Ni}^{2+} / \text{Ni}) = \varphi^\ominus(\text{Ni}^{2+} / \text{Ni}) + \frac{2.303RT}{2F} \lg \{c(\text{Ni}^{2+}) / c^\ominus\}$$

$$= -0.25\text{V} + \frac{0.059\text{V}}{2} \times \lg 0.080 = -0.28\text{V}$$

$$\varphi(\text{Zn}^{2+} / \text{Zn}) = \varphi^\ominus(\text{Zn}^{2+} / \text{Zn}) + \frac{2.303RT}{2F} \lg \{c(\text{Zn}^{2+}) / c^\ominus\}$$

$$= -0.76\text{V} + \frac{0.059\text{V}}{2} \times \lg 0.020 = -0.81\text{V}$$

$$\varepsilon = -0.28\text{V} - (-0.81\text{V}) = 0.53\text{V}$$

(2) $\text{Cl}^-(1.0\text{mol/L}) | \text{Cl}_2(100\text{kPa}) || \text{Fe}^{3+}(0.10\text{mol/L}), \text{Fe}^{2+}(1.0\text{mol/L})$

$$\varphi(\text{Fe}^{3+} / \text{Fe}^{2+}) = \varphi^\ominus(\text{Fe}^{3+} / \text{Fe}^{2+}) + \frac{2.303RT}{F} \lg \frac{\{c(\text{Fe}^{3+}) / c^\ominus\}}{\{c(\text{Fe}^{2+}) / c^\ominus\}}$$

$$= 0.77\text{V} + 0.059\text{V} \times \lg \frac{0.10}{1.0} = 0.71\text{V}$$

$$\varepsilon = 1.36\text{V} - 0.71\text{V} = 0.65\text{V}$$

(3) $\text{Cl}^-(1.0\text{mol/L}) | \text{Cl}_2(100\text{kPa}) || \text{Cr}_2\text{O}_4^{2-}(1.0\text{mol/L}), \text{H}^+(10\text{mol/L}), \text{Cr}^{3+}(1.0\text{mol/L})$

$$\begin{aligned}\varphi(\text{Cr}_2\text{O}_7^{2-}/\text{Cr}^{3+}) &= \varphi^\ominus(\text{Cr}_2\text{O}_7^{2-}/\text{Cr}^{3+}) + \frac{2.303RT}{6F} \lg \frac{\{c(\text{Cr}_2\text{O}_7^{2-})/c^\ominus\}\{c(\text{H}^+)/c^\ominus\}^{14}}{\{c(\text{Cr}^{3+})/c^\ominus\}^2} \\ &= 1.23\text{V} + \frac{0.059\text{V}}{6} \times \lg 10^{14} = 1.37\text{V} \\ \varphi(\text{Cl}_2/\text{Cl}^-) &= \varphi^\ominus(\text{Cl}_2/\text{Cl}^-) = 1.36\text{V} \\ \varepsilon &= 0.01\text{V}\end{aligned}$$

(结果说明, 在强酸性介质中, 重铬酸根亦可以氧化氯离子)

9.11 (1)

$$\begin{aligned}\varphi^\ominus(\text{Sn}^{2+}/\text{Sn}) &= -0.138\text{V} \\ \varphi^\ominus(\text{Pb}^{2+}/\text{Pb}) &= -0.126\text{V} \\ \Delta_r G_m^\ominus &= -2F\{-0.138\text{V} - (-0.126\text{V})\} = 2.32 \times 10^3 \text{J/mol} \\ \varphi(\text{Sn}^{2+}/\text{Sn}) &= \varphi^\ominus(\text{Sn}^{2+}/\text{Sn}) + \frac{2.303RT}{2F} \lg \{c(\text{Sn}^{2+})/c^\ominus\} \\ &= -0.138\text{V} + \frac{0.059\text{V}}{2} \times \lg 0.050 = -0.176\text{V} \\ \varphi(\text{Pb}^{2+}/\text{Pb}) &= \varphi^\ominus(\text{Pb}^{2+}/\text{Pb}) + \frac{2.303RT}{2F} \lg \{c(\text{Pb}^{2+})/c^\ominus\} \\ &= -0.126\text{V} + \frac{0.059\text{V}}{2} \times \lg 0.50 = -0.135\text{V}\end{aligned}$$

$$\Delta_r G_m = -2F\{-0.176\text{V} - (-0.135\text{V})\} = 7.91 \times 10^3 \text{J/mol}$$

$$\begin{aligned}\lg K^\ominus &= \frac{2F\{\varphi^\ominus(\text{Sn}^{2+}/\text{Sn}) - \varphi^\ominus(\text{Pb}^{2+}/\text{Pb})\}}{2.303RT} \\ &= \frac{2 \times \{-0.138\text{V} - (-0.126\text{V})\}}{0.059\text{V}} = -0.407\end{aligned}$$

$$K^\ominus = 0.39$$

因为: $\varphi(\text{Sn}^{2+}/\text{Sn}) < \varphi(\text{Pb}^{2+}/\text{Pb})$

或因: $\Delta_r G_m > 0$

或因: $Q = 10 > K^\ominus$

所以反应逆向自发进行。

(2)

$$\begin{aligned}\varphi^\ominus(\text{NO}_3^-/\text{NO}) &= 0.957\text{V} \\ \varphi^\ominus(\text{Fe}^{3+}/\text{Fe}^{2+}) &= 0.771\text{V} \\ \Delta_r G_m^\ominus &= -3F\{\varphi^\ominus(\text{NO}_3^-/\text{NO}) - \varphi^\ominus(\text{Fe}^{3+}/\text{Fe}^{2+})\} = \\ &= -3 \times 96500 \text{C/mol} \times \{0.957\text{V} - 0.771\text{V}\} = -5.38 \times 10^4 \text{J/mol}\end{aligned}$$

$$\begin{aligned}\varphi(\text{NO}_3^-/\text{NO}) &= \varphi^\ominus(\text{NO}_3^-/\text{NO}) + \frac{2.303RT}{3F} \lg \frac{\{c(\text{NO}_3^-)/c^\ominus\}\{c(\text{H}^+)/c^\ominus\}^4}{\{p(\text{NO})/p^\ominus\}} \\ &= 0.957\text{V} + \frac{0.059\text{V}}{3} \times \lg(1.0 \times 10^{-3})^4 = 0.721\text{V}\end{aligned}$$

$$\begin{aligned}\varphi(\text{Fe}^{3+}/\text{Fe}^{2+}) &= \varphi^\ominus(\text{Fe}^{3+}/\text{Fe}) \\ &= 0.771\text{V}\end{aligned}$$

$$\Delta_r G_m = -3F\{\varphi(\text{NO}_3^-/\text{NO}) - \varphi(\text{Fe}^{3+}/\text{Fe}^{2+})\} = -3 \times 96500\text{C/mol} \times \{0.721\text{V} - 0.771\text{V}\} = 1.45 \times 10^4\text{J/mol}$$

$$\begin{aligned}\lg K^\ominus &= \frac{3F\{\varphi^\ominus(\text{NO}_3^-/\text{NO}) - \varphi^\ominus(\text{Fe}^{3+}/\text{Fe}^{2+})\}}{2.303RT} \\ &= \frac{3 \times \{0.957\text{V} - 0.771\text{V}\}}{0.059\text{V}} = 9.46 \\ K^\ominus &= 2.87 \times 10^9\end{aligned}$$

因为: $\varphi(\text{NO}_3^-/\text{NO}) < \varphi(\text{Fe}^{3+}/\text{Fe}^{2+})$

或因: $\Delta_r G_m > 0$

或因: $Q = 1.0 \times 10^{12} > K^\ominus$

所以反应逆向自发进行

结果说明, 定性分析中利用棕色环法检验 NO_3^- , 若在 pH 约等于 3 的醋酸介质中, 反应不能进行。该反应应在浓硫酸介质中进行。

(3)

$$\begin{aligned}\varphi^\ominus(\text{HNO}_2/\text{NO}) &= 0.983\text{V} \\ \varphi^\ominus(\text{Fe}^{3+}/\text{Fe}^{2+}) &= 0.771\text{V}\end{aligned}$$

$$\Delta_r G_m^\ominus = -F\{\varphi^\ominus(\text{HNO}_2/\text{NO}) - \varphi^\ominus(\text{Fe}^{3+}/\text{Fe}^{2+})\} = -96500\text{C/mol} \times \{0.983\text{V} - 0.771\text{V}\} = -2.05 \times 10^4\text{J/mol}$$

$$\begin{aligned}\varphi(\text{HNO}_2/\text{NO}) &= \varphi^\ominus(\text{HNO}_2/\text{NO}) + \frac{2.303RT}{F} \lg \frac{\{c(\text{HNO}_2)/c^\ominus\}\{c(\text{H}^+)/c^\ominus\}}{\{p(\text{NO})/p^\ominus\}} \\ &= 0.983\text{V} + 0.059\text{V} \times \lg(1.0 \times 10^{-3}) = 0.806\text{V} \\ \varphi(\text{Fe}^{3+}/\text{Fe}^{2+}) &= \varphi^\ominus(\text{Fe}^{3+}/\text{Fe}) \\ &= 0.771\text{V}\end{aligned}$$

$$\Delta_r G_m = -F\{\varphi(\text{HNO}_2/\text{NO}) - \varphi(\text{Fe}^{3+}/\text{Fe}^{2+})\} = -96500\text{C/mol} \times \{0.806\text{V} - 0.771\text{V}\} = -3.38 \times 10^3\text{J/mol}$$

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所以可利用反应 $\text{CuCl}_2 + \text{Cu} = 2\text{CuCl}$ 制备 CuCl 。

9.15

答案：若 (1) 为正极， $c(\text{H}^+) = 0.054\text{mol/L}$

若 (2) 为正极， $c(\text{H}^+) = 0.187\text{mol/L}$

9.16

$$\varphi^\ominus(\text{O}_2/\text{OH}^-) = 1.229\text{V} + \frac{0.059\text{V}}{4} \times \lg(1.0 \times 10^{-14})^4 = 0.403\text{V}$$

(所求实际为 $c(\text{H}^+) = 10^{-14}\text{mol/L}$ 时， $\text{O}_2/\text{H}_2\text{O}$ 电极的电极电势)

9.17

$$\varphi^\ominus(\text{Hg}_2\text{Cl}_2/\text{Hg}) = 0.281\text{V}$$

$$\text{所以：}\varphi(\text{H}^+/\text{H}_2) = \varphi^\ominus(\text{Hg}_2\text{Cl}_2/\text{Hg}) - \varepsilon = 0.281\text{V} - 0.48\text{V} = -0.20\text{V}$$

$$\varphi(\text{H}^+/\text{H}_2) = \varphi^\ominus(\text{H}^+/\text{H}_2) + \frac{2.303RT}{2F} \lg \frac{\{c(\text{H}^+)/c^\ominus\}^2}{p(\text{H}_2)/p^\ominus}$$

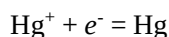
$$\therefore 0.20\text{V} = 0.059\text{V} \times \lg\{c(\text{H}^+)/c^\ominus\}$$

$$c(\text{H}^+) = 4.1 \times 10^{-4}\text{mol/L}$$

$$\therefore c(\text{H}^+)/c^\ominus = \sqrt{K_a^\ominus \times c(\text{HB})/c^\ominus}$$

$$\therefore K_a^\ominus = 1.7 \times 10^{-6}$$

9.18 若亚汞离子为 Hg^+ ，则电极反应为：



根据此电极反应，可写出二电极的 Nernst 方程并可计算电池电动势：

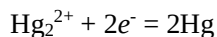
$$\varphi(+) = \varphi^\ominus + \frac{2.303RT}{F} \lg\{c(\text{Hg}^+)/c^\ominus\}$$

$$\varphi(-) = \varphi^\ominus + \frac{2.303RT}{F} \lg\{c(\text{Hg}^+)/10c^\ominus\}$$

$$\varepsilon = \varphi(+) - \varphi(-) = \frac{2.303RT}{F} \lg 10 = 0.059\text{V}$$

计算结果与实验数据不符。

若亚汞离子为 Hg_2^{2+} ，则电极反应为：



根据此电极反应，可写出二电极的 Nernst 方程并可计算电池电动势：

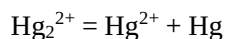
$$\varphi(+) = \varphi^\ominus + \frac{2.303RT}{2F} \lg\{c(\text{Hg}_2^{2+})/c^\ominus\}$$

$$\varphi(-) = \varphi^\ominus + \frac{2.303RT}{2F} \lg\{c(\text{Hg}_2^{2+})/10c^\ominus\}$$

$$\varepsilon = \varphi(+) - \varphi(-) = \frac{2.303RT}{2F} \lg 10 = 0.03\text{V}$$

计算结果与实验数据相符，故可知亚汞离子为 Hg_2^{2+} 。

9.19



$$\lg K^\ominus = \frac{F\{\varphi^\ominus(\text{Hg}_2^{2+}/\text{Hg}) - \varphi^\ominus(\text{Hg}^{2+}/\text{Hg}_2^{2+})\}}{2.303RT} = -2.034$$

$$K^\ominus = 9.2 \times 10^{-3}$$

$$K^\ominus = \frac{c(\text{Hg}^{2+})/c^\ominus}{c(\text{Hg}_2^{2+})/c^\ominus}$$

$$\therefore \frac{c(\text{Hg}^{2+})}{c(\text{Hg}_2^{2+})} = \frac{1}{110}$$

9.20

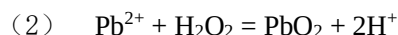
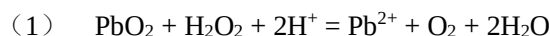
根据标准电极电势图:

$$\varphi^\ominus(\text{PbO}_2/\text{Pb}^{2+}) = 1.455\text{V}$$

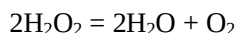
$$\varphi^\ominus(\text{O}_2/\text{H}_2\text{O}_2) = 0.695\text{V}$$

$$\varphi^\ominus(\text{H}_2\text{O}_2/\text{H}_2\text{O}) = 1.76\text{V}$$

下列反应可自发进行:

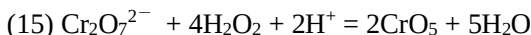
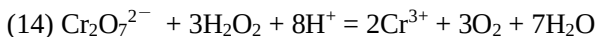
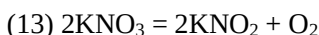
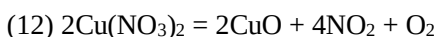
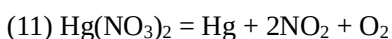
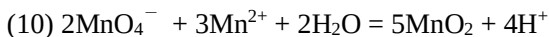
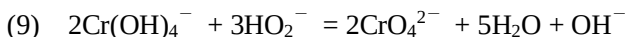
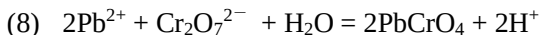
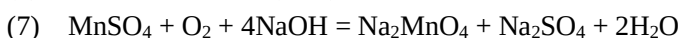
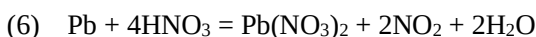
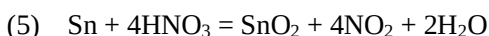
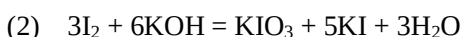
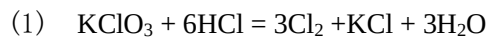


(1) + (2):



故 PbO_2 (MnO_2 等) 可催化过氧化氢歧化分解。

9.22



第十章 配位化合物

10.2



10.3 (2)

10.4 (4)

10.5



10.6 (4)

10.7 (3)

10.8 (3)

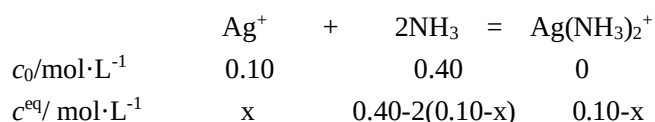
10.9 (1)

10.10 (4)

10.11 (3)

10.12 (1)

10.13



$$K_f^\ominus\{\text{Ag}(\text{NH}_3)_2^+\} = \frac{0.10}{x \times 0.20^2} = 1.1 \times 10^7$$

$$x = 2.3 \times 10^{-7}$$

$$\therefore c(\text{Ag}^+) = 2.3 \times 10^{-7} \text{ mol}\cdot\text{L}^{-1}$$

10.14



$$K_f^\ominus\{\text{Ag}(\text{NH}_3)_2^+\} = \frac{0.099}{0.0010x^2} = 1.1 \times 10^7$$

$$x = 3.0 \times 10^{-3}$$

$$\therefore c(\text{NH}_3) = 3.0 \times 10^{-3} \text{ mol}\cdot\text{L}^{-1}$$

$$c(\text{NH}_4^+) = 0.10 \text{ mol}\cdot\text{L}^{-1} + 0.0020 \text{ mol}\cdot\text{L}^{-1} - 3.0 \times 10^{-3} \text{ mol}\cdot\text{L}^{-1} = 0.099 \text{ mol}\cdot\text{L}^{-1}$$

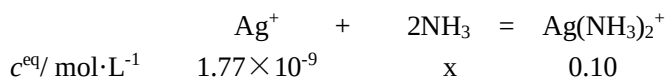
$$\text{pOH} = \text{p}K_b^\ominus(\text{NH}_3) - \lg \frac{c(\text{NH}_3)}{c(\text{NH}_4^+)} = 4.76 - 1.52 = 6.24$$

$$\text{pH} = 7.76$$

10.15

开始生成沉淀时:

$$c(\text{Ag}^+)/c^\ominus = \frac{K_{\text{sp}}^\ominus(\text{AgCl})}{c(\text{Cl}^-)/c^\ominus} = \frac{1.77 \times 10^{-10}}{0.10} = 1.77 \times 10^{-9}$$



$$K_f^{\ominus}\{\text{Ag}(\text{NH}_3)_2^+\} = \frac{0.10}{1.77\times 10^{-9} \times x^2} = 1.1\times 10^7$$

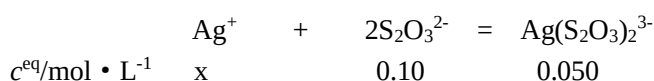
$$x = 2.3$$

$$\therefore c(\text{NH}_3) = 2.3\text{mol}\cdot\text{L}^{-1}$$

$$c(\text{NH}_4^+) = 5.0\text{mol}\cdot\text{L}^{-1} - 2.3\text{mol}\cdot\text{L}^{-1} = 2.7\text{mol}\cdot\text{L}^{-1}$$

$$\text{pOH} = \text{p}K_b^{\ominus}(\text{NH}_3) - \lg \frac{c(\text{NH}_3)}{c(\text{NH}_4^+)} = 4.82 \quad \text{pH} = 9.18$$

10.16



$$K_f^{\ominus} = \frac{0.050}{x \times 0.10^2} = 2.9 \times 10^{13} \quad x = 1.7 \times 10^{-13}$$

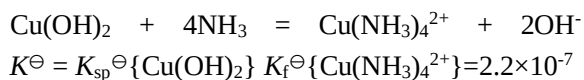
$$\text{即: } c(\text{Ag}^+) = 1.7 \times 10^{-13} \text{mol}\cdot\text{L}^{-1}$$

欲生成沉淀所需 KCl 和 KI 的浓度分别为:

$$c(\text{KCl}) = \frac{2c^{\ominus} \times K_{\text{sp}}^{\ominus}(\text{AgCl})}{c(\text{Ag}^+)/c^{\ominus}} = 2.0 \times 10^3 \text{mol}\cdot\text{L}^{-1}$$

$$c(\text{KI}) = \frac{2c^{\ominus} \times K_{\text{sp}}^{\ominus}(\text{AgI})}{c(\text{Ag}^+)/c^{\ominus}} = 1.0 \times 10^{-3} \text{mol}\cdot\text{L}^{-1}$$

10.17



反应难于自发进行

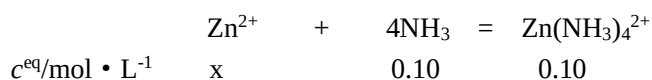


$$K^{\ominus} = K_{\text{sp}}^{\ominus}\{\text{Cu}(\text{OH})_2\} K_f^{\ominus}\{\text{Cu}(\text{NH}_3)_4^{2+}\} \{1/K_b^{\ominus}(\text{NH}_3)\}^2 = 1.4 \times 10^3$$

反应易于自发进行, 原因为 NH_4^+ 与生成的 OH^- 结合为 NH_3 , 化学平衡正向移动。

10.18 不能, Co^{3+} , +3, 小于, 大于

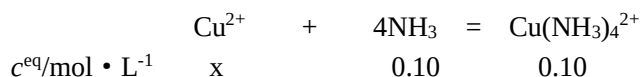
10.19



$$K_f^{\ominus}\{\text{Zn}(\text{NH}_3)_4^{2+}\} = \frac{0.10}{x \times 0.10^4} = 2.9 \times 10^9$$

$$c(\text{Zn}^{2+})/c^{\ominus} = x = 3.4 \times 10^{-7}$$

$$\varphi(\text{Zn}^{2+}/\text{Zn}) = \varphi^{\ominus}(\text{Zn}^{2+}/\text{Zn}) + \frac{2.303RT}{2F} \lg\{c(\text{Zn}^{2+})/c^{\ominus}\} = -0.95\text{V}$$



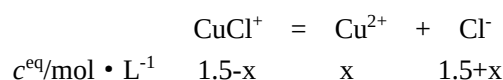
$$K_f^{\ominus}\{\text{Cu}(\text{NH}_3)_4^{2+}\} = \frac{0.10}{x \times 0.10^4} = 2.1 \times 10^{13}$$

$$c(\text{Cu}^{2+})/c^{\ominus} = x = 4.8 \times 10^{-11}$$

$$\varphi(\text{Cu}^{2+}/\text{Cu}) = \varphi(\text{Cu}^{2+}/\text{Cu}) + \frac{2.303RT}{2F} \lg\{c(\text{Cu}^{2+})/c^{\ominus}\} = 0.04\text{V}$$

$$\therefore \varepsilon = 0.04\text{V} - (-0.95\text{V}) = 0.99\text{V}$$

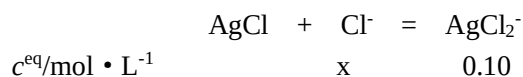
10.20



$$K_f^{\ominus} = \frac{1.5-x}{x(1.5+x)} = 1.0 \quad x = 0.50$$

$$c(\text{CuCl}^+) = 1.0 \text{ mol} \cdot \text{L}^{-1} \quad c(\text{Cu}^{2+}) = 0.50 \text{ mol} \cdot \text{L}^{-1}$$

10.21



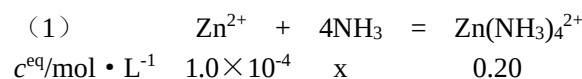
$$K^{\ominus} = K_{\text{sp}}^{\ominus}(\text{AgCl})K_f^{\ominus}(\text{AgCl}_2^-) = \frac{0.10}{x} = 5.3 \times 10^{-5}$$

$$x = 1.9 \times 10^3$$

$$m(\text{NaCl}) = c(\text{NaCl})VM(\text{NaCl}) = 1.1 \times 10^4 \text{g}$$

加入沉淀剂过多，可能由于配合物的生成反而使沉淀溶解度增大。

10.22



$$K_f^{\ominus} = \frac{0.20}{1.0 \times 10^{-4} \times x^4} = 2.9 \times 10^9$$

$$x = 0.029$$

$$\text{即: } c(\text{NH}_3) = 0.029 \text{ mol} \cdot \text{L}^{-1}$$

$$c(\text{OH}^-)/c^{\ominus} = \sqrt{K_b^{\ominus}(\text{NH}_3)c(\text{NH}_3)/c^{\ominus}} = 7.2 \times 10^{-4}$$

$$\{c(\text{Zn}^{2+})/c^{\ominus}\}\{c(\text{OH}^-)/c^{\ominus}\}^2 = 5.2 \times 10^{-11} > K_{\text{sp}}^{\ominus}\{\text{Zn}(\text{OH})_2\}$$

$\therefore$ 有沉淀生成

(2) 同理可计算得，当 $c(\text{Zn}^{2+}) = 1.0 \times 10^{-15} \text{ mol} \cdot \text{L}^{-1}$ 时：

$$c(\text{NH}_3) = 16 \text{ mol} \cdot \text{L}^{-1}$$

$$c(\text{OH}^-) = 0.017 \text{ mol} \cdot \text{L}^{-1}$$

$$\{c(\text{Zn}^{2+})/c^{\ominus}\}\{c(\text{OH}^-)/c^{\ominus}\}^2 = 2.9 \times 10^{-19} < K_{\text{sp}}^{\ominus}\{\text{Zn}(\text{OH})_2\}$$

$\therefore$ 无沉淀生成

向含有 Zn^{2+} 离子的溶液中滴加氨水，开始有白色 $\text{Zn}(\text{OH})_2$ 沉淀生成，继而沉淀溶解，得到无色的 $\text{Zn}(\text{NH}_3)_4^{2+}$ 溶液。